

Firm-level price wedges for pricewedge.com

Three design decisions and the evidence behind them

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Summary

The website reports a price wedge for an individual firm. Rebuilding the paper's construction from raw CRSP and Compustat data exposed three choices that the paper does not discuss and that move a firm's number by ten to twenty percentage points (pp). This note records what was chosen and why.

1. **What the firm-level number means.** It is the price wedge that firms with the firm's characteristics realized, each firm counting once, with the value-weighted average of all firm-level wedges held at the market's wedge of zero. The paper's published firm-level wedges do not have this property: they run about 18 pp too high for the typical firm, because the mapping describes each portfolio by its average *member* while its wedge belongs to its *dollars*. Section 2.
2. **How the characteristics are defined.** As the paper the anomaly comes from defines them. Six of the 57 characteristics in the paper's data depart from their sources; five are corrected and one is dropped. Section 3.
3. **Which sample.** All available formation dates (July 1964 to January 2011, cash flows to December 2025), and no alternative samples on the website. Section 4.

The evidence is in two companion documents: the quality report (the mapping and the level tests) and the signal log (every characteristic's construction and its check against the paper's own data). Every number here is produced by a script in the pipeline and can be regenerated.

1. The price wedge, and the one calibration that defines it

The price wedge is the log deviation of a stock's market price from its fundamental value, the present value of its future dividends under a benchmark stochastic discount factor (SDF). For a portfolio formed at date t it is estimated as

$$PW_t = -\log \mathbb{E} \left[\frac{\sum_{s=1}^{180} m_{t,t+s} D_{t+s} + m_{t,t+180} P_{t+180}}{P_t} \right],$$

the expectation being the average over formation dates, one per month. A negative wedge means the price is below fundamental value. The SDF is the CAPM-based one of the paper, with one free parameter, the market price of risk, chosen so that **the price wedge of the value-weighted market portfolio is zero**. Every wedge is therefore relative to the average dollar in the market. This calibration is kept unchanged throughout.

Two facts about the object matter below.

A portfolio's wedge is a dollar-weighted quantity. It is the wedge of one dollar spread across the portfolio's members in proportion to their market capitalization, so a few large members set it.

The average firm is far from the average dollar. With the paper's price of risk, the value-weighted market has a wedge of zero by construction; the equal-weighted market, one dollar in every firm, has a realized wedge of -24.6 pp. The average firm is a small firm, and under this SDF small firms are strongly underpriced. Any firm-level number has to say which of the two averages it refers to.

2. Decision 1: what the firm-level price wedge means

How the paper carries portfolio wedges to firms

Each of the 570 characteristic-sorted decile portfolios has a price wedge. Each also has a set of *portfolio-level characteristics*: the rank-normalized characteristics of its members, averaged over the members and over time. A regression of the wedges on the portfolio-level characteristics gives a rule from characteristics to wedge, and evaluating that rule at an individual firm's own characteristics gives the firm-level price wedge. The paper uses three principal components of the characteristics of the 114 extreme deciles; the pipeline uses a penalized regression on all 570, which fits better on every portfolio-level test (quality report, Section 1). That difference is not the decision here.

The decision is hidden in the word *average*. A portfolio's average characteristics can be taken over its members with equal weights, which describes its typical member, or with market-capitalization weights, which describes its typical dollar. For the value-weighted book-to-market decile 10, the typical member has a size rank near 0.4 and the typical dollar a size rank near 0.98. The paper's code uses equal weights (with the comment "SHOULD THIS BE VALUE WEIGHTED AVERAGE?????" beside it); the pipeline had used capitalization weights. The choice moves every firm-level wedge by ten to twenty-seven points, and it is the whole reason the pipeline's firm-level wedges first sat eleven points below the published ones: with equal weights the published firm-level wedges are reproduced to 1.2 pp (slope 1.04, correlation 0.99).

The test

Two firm-level series can correlate at 0.95 and differ by 20 pp in level, and the level is what the website prints. The test used here therefore compares assigned wedges with realized ones. Take a candidate firm-level wedge; sort firms into deciles on it every month, as one would on a characteristic; hold each decile for fifteen years, equal-weighted and value-weighted; compute the decile's realized price wedge with the same SDF; and regress the realized wedge on the average wedge the candidate had assigned to the decile's members. A candidate whose levels can be trusted gives a slope of one and an intercept of zero. The equal-weighted deciles answer the website's question, whether firms realize what they are told; the value-weighted deciles check that the candidate respects the calibration. The calibration gives one more check: the value-weighted average of all firm-level wedges must equal the market's wedge, zero.

Three candidates

Table 1. Three candidates for the firm-level price wedge, and what the level test says about each.

Candidate	What the number claims	equal-weighted deciles		value-weighted	
		slope	intercept	slope	VW mean
1. Published construction	“the number the paper’s method assigns”	0.64	−18	0.95	+9
2. Per dollar	“a dollar in firms like this one is mispriced by X ”	0.67	−3	0.79	+1
3. Per firm, aggregate pinned	“firms like this one are mispriced by X on average; dollar-weighted, mispricing is zero”	0.86	−1	0.76	0

Candidate 1: value-weighted portfolio wedges regressed on equal-weighted portfolio characteristics. Candidate 2: value-weighted wedges and characteristics. Candidate 3: value- and equal-weighted portfolio sets together, each with characteristics of its own weighting. Intercepts in pp. “VW mean” is the value-weighted average of the candidate’s firm-level wedges, which should be near zero.

Candidate 1 learns “a portfolio whose typical member is a small value firm has wedge W ” when W was earned by the portfolio’s large members, and then hands the small firm the large firm’s wedge. Firms it calls fairly priced realize about −25 pp; only the most underpriced decile realizes what it is assigned.

Candidate 2 is internally consistent but is evaluated far outside the points it was fitted on: almost every portfolio’s dollar-average size rank is above 0.85, and the website asks about firms at 0.4. The level is right on average and the extremes are exaggerated (the most underpriced decile is assigned −70 pp and realizes −42).

Candidate 3 adds the same 570 deciles held with equal weights, whose wedge is the wedge of the average member, each described by the equal-weighted average of its members’ characteristics. A small firm’s number is then interpolated from portfolios whose members resemble it, and the value-weighted portfolios keep the dollar-weighted total anchored. Out of sample (mapping fitted on formation dates before 1988, evaluated after) the slopes hold at 0.87 to 0.99.

Decision

Candidate 3 is adopted. It is the only candidate whose levels were checked against realized outcomes at the firm level and found right, equal- and value-weighted at once; its claim is the one a reader looking up a single company expects; it keeps the paper’s calibration intact; and it changes rankings very little (all candidates rank firms alike, correlations 0.9 and above). The published construction is kept as a labelled reference series.

A consequence for the paper. The published firm-level wedges are candidate 1, so any use of them in levels inherits the −18 pp intercept; the investment- q regressions use them as a regressor in levels. Rankings are unaffected.

3. Decision 2: how the characteristics are defined

The mapping learns from the wedges of the 570 portfolios, and a portfolio is defined by how its sorting characteristic was built. Rebuilding the 57 characteristics and checking them against the paper's own firm-month panels verified 56 of them and found six places where what the paper's data does differs from the source the paper cites. The rule adopted: **each characteristic is defined as the paper the anomaly comes from defines it**, read through Freyberger, Neuhierl and Weber (2020), whose list the paper takes its characteristics from. The point of the anomalies is to use the existing specifications, not new ones that arose from an implementation slip.

Table 2. The six characteristics whose construction in the paper's data departs from its source.

	Source	Paper's data	Website
OA, AOA	Sloan (1996): accruals <i>net of</i> depreciation	depreciation added	Sloan's sign, lagged assets as in FNW
SUV	Garfinkel (2009): a daily quantity, the day's volume minus a regression fit from a prior window, over the fit's residual s.d.; FNW: fit on the previous month	the last trading day of a regression run within the same month	daily values from the previous month's fit, averaged over the month
DTO	Garfinkel (2009): daily turnover minus market turnover, minus a control-period mean; FNW: minus the firm's 180-day median, Nasdaq volume scaled (Anderson and Dyl 2005)	last day's raw turnover minus its 180-day mean; no market adjustment, no scaling	FNW's definition, averaged over the month
TNOVR	Datar, Naik, Radcliffe (1998); FNW: last month's volume over shares	last month's volume over shares	unchanged (the data follows its source)
aBEME, aSIZE, aSAT	Asness, Porter, Stevens (2000): 48 Fama-French industries by SIC, equal-weighted industry mean, three firms or more; FNW apply the same to PM and SAT	27 industry groups of unknown origin	Fama-French 48, equal-weighted mean
aPM	Soliman (2008) defines PM; the adjustment is FNW's	industry means not reconstructible; long-short alpha zero	dropped

Two remarks. For SUV and DTO neither source singles out one trading day, so the paper's last-day values are a construction artifact (the signature of a daily series collapsed to monthly by keeping the last observation); the monthly average is Garfinkel's own aggregation over his event window. Soliman's published paper does not industry-adjust; the adjustment of PM and SAT is FNW's, built like Asness, Porter and Stevens's, and that is what is used. Implementation conventions that the sources leave open and that do not change what a characteristic measures (the 249-day window of the daily beta, zero-volume days excluded from the volume dispersion, the day-level bid-ask

spread, dividends in dollars over market capitalization) are kept and documented in `signals.pdf`.

4. Decision 3: which formation dates

The wedges, and the price of risk with them, are averages over formation dates. Three candidates: the paper's window (July 1964 to December 2002, cash flows to 2017), all available dates (to January 2011, cash flows to 2025), or a later window.

Extending the paper's window to all available dates changes almost nothing: the two cross-sections of 560 portfolio wedges correlate at 0.99, one long-short spread changes sign, firm-level wedges move by 2 pp. Dropping the early dates is different. The first and second halves of the sample (1964–87 and 1988–2011) correlate at 0.45, where two halves of a constant cross-section would correlate at about 0.68 given their estimation noise; the value ratios reverse sign between halves, profitability spreads shrink, momentum grows. Any window starting in 1980 or later rests on one or two fifteen-year cycles (the price of risk swings from 5.6 to 2.5 to 2.8 across thirds of the sample) and moves the typical firm's wedge by 10 to 20 pp. Firm rankings are robust to the choice; levels are not.

Decision. All available formation dates. A later window is not a better estimate of the same quantity but an estimate of a different regime. The half-sample estimates stay as an internal check on every build; the website offers no sample choices, because that would present regime dependence as if it were an alternative estimate.

5. What the website reports, and what every build must pass

Reported per firm: the candidate-3 price wedge on all available dates with the characteristics of Section 3, the firm's percentile among all firms that month, and a flag when the firm's characteristics lie outside the range spanned by the portfolios (where the mapping extrapolates). The published construction is available as a labelled reference. Uncertainty is stated once, on the method page: about 30 pp common to all firms (the market's realized path), about 3 to 4 pp specific to a firm, and about ± 12 pp of sensitivity to the half of the sample used.

Acceptance tests, all on the latest build:

1. Replication: with the paper's own definitions, at least 56 of 57 characteristics agree with the paper's panels and Table 1 within a point on both legs.
2. Levels: on equal-weighted deciles of the assigned wedge, slope at least 0.85 and intercept within 3 pp; on value-weighted deciles, slope at least 0.75; the decile table reported.
3. Aggregate: the value-weighted average of firm-level wedges within a point or two of zero.
4. Out-of-sample fit of the mapping (leave one family of characteristics out) reported and not lower than the previous build's by more than noise.
5. Subsequent returns of the assigned-wedge deciles monotone at five years.
6. Half-sample check: the mappings fitted on 1964–87 and on 1988–2011 and their firm-level difference from the headline reported.

Appendix: scripts and sources

Table 3. Where each piece lives.

build_panel.py --spec site	the characteristics as their sources define them, to 2025
portfolio_wedges.py,	sorts, calibration, wedges and portfolio-level characteris-
portfolio_profiles.py	tics, value- and equal-weighted (--weighting equal)
firm_wedges.py --mapping	the candidate-3 mapping
stacked	
realised_levels.py	the level test and the aggregate check
stability.py	wedges on subsets of formation dates
validate.py, compare_panels.py	replication against Table 1 and the paper's panels
quality report, signal log	the evidence (doc/quality.pdf, doc/signals.pdf)
Literature folder	the source papers (Website/Literature/, indexed in _index.md)